Calculations report: TEST - verification

Job number: TEST-001

Customer: TEST

Revision A, 2026-10-23

NOT APPROVED: issued for review only.

The design rules are this software's defaults; the company has not confirmed them.

Basis of calculation

- Design current Ib = P / (k U cos phi); breaker In >= Ib, cable Iz >= In (IEC 60364-4-43 §433.1; ABB Electrical Installation Handbook Vol. 2, §2.1-2.3).
- Current-carrying capacity from IEC 60364-5-52 Annex B, corrected for ambient temperature and grouping.
- Voltage drop from the origin, from Schneider EIG Fig. G28 (copper) and ABB §2.2.2 (aluminium), against IEC 60364-5-52 Annex G or the company's limits.
- Fault level at a sub-board through its feeder, IEC 60909-0 (cmax 1.10); breaking capacity at or above it.
- Earth fault disconnection, IEC 60364-4-41 §411.4.4: Zs x Ia <= 0.95 U0, Ia the top of the breaker's instantaneous band (IEC 60898-1).
- Short circuit along the cable, ABB §2.4: the far-end Ikmin trips the breaker at once; k^2S^2 from Table 1 for comparison with the breaker's let-through I^2t.

Board DB-POOL

400 V, 3-phase, TN-S · Ik - kA · Ze - ohm · Incomer C63 4P

Circuit	Ib A	Protection	Cable	L m	dU %	Zs ohm	Zs max	k^2S^2 $(kA)^2s$
Pool underwater lighting (TEST VALUE)	1.9	C10 1P	3G1.5 Cu PVC	20	0.5	-	2.194	0.03
Pool heater (TEST VALUE)	9.6	C10 3P	5G1.5 Cu PVC	15	0.9	-	2.194	0.03
Plant room sockets (TEST VALUE)	7.2	C16 1P	3G2.5 Cu PVC	10	0.56	-	1.372	0.083
Pool filtration pump (TEST VALUE)	5	MF5 3P	4G1.5 Cu PVC	15	0.47	-	-	0.03

Notes

- Group 1 breaker raised to 32 A, 1.6 x the 16 A breaker after it, for discrimination.
- Pool filtration pump (TEST VALUE): Ir 5 A, the typical Ir of a 2.2 kW motor in ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3; check it against the nameplate.
- Pool filtration pump (TEST VALUE): dol starter, Type 2 coordination, ABB Electrical installation handbook Vol. 2 (1SDC010001D0204), §3.3 Table 3: 400 V 50 kA DOL Normal Type 2.
- Phase imbalance 33.1 % exceeds the company's 10 %: the loads cannot be spread more evenly as given.
- Incomer raised to 63 A, 1.6 x the 32 A breaker after it, for discrimination.

- No fault level given: every breaker's breaking capacity is left to be confirmed.

- Leave 1 spare outgoing ways (20 %).

- Loads without a power factor were taken at cos phi 0.9.

- Each breaker is rated at least 1.6 x the largest breaker after it, so they discriminate on overload. On short circuit, miniature breakers discriminate only up to the upstream breaker's instantaneous trip (about 5 x In for curve C); where total selectivity is required, confirm it from the manufacturer's selectivity tables.

- No Ze is given for this board, so circuits not under a residual current device are not checked for disconnection on an earth fault; enter the measured or declared Ze.

- Cables sized for method B1, 30 °C, 1 grouped circuit(s), 70 °C insulation.

- Circuit rules are this software's defaults, not confirmed by the company's engineers.

- Enclosure: none named, since 3 devices have no width (rail_widths_mm in the company settings).

Prepared	Checked	Approved
V4		